SUBJECT CODE : 23A01702b

RAILWAYS, AIRPORTS, DOCKS AND HARBOUR ENGINEERING

UNIT – V : PORTS AND HARBOURS

(Handwritten Study Notes - Exam Oriented)

Syllabus (as given):

Harbours – Requirements of Ports and Harbors – Types of Ports – Classification of Harbors – Docks and Types of Docks – Dry Docks, Wharves and Jetties – Breakwaters: Layouts of Different Types of Harbors and Docks – Dredging Operations – Navigation Aids

1. Introduction: Harbours and Ports

A **harbour** is a **sheltered water area** (protected from waves, wind and currents) in which ships can **anchor, load/unload, take shelter and be repaired** safely. A **port** is a harbour **plus** the facilities for transfer of cargo and passengers between sea and land: quays, cranes, warehouses, rail/road links, customs and so on.

- Water transport is the **cheapest** mode for bulk and international trade; about 90 % of world trade (by volume) moves by sea; India has a coastline of about 7500 km.
- **Port = harbour + land-side facilities** (berths, transit sheds, storage yards, container terminals, cranes, tugs, pilotage).
- Ports are managed in India by the **Ministry of Ports, Shipping and Waterways**; 12 **major ports** are under the Central Government and many **minor / intermediate ports** under the State Governments.

Requirements of a good port / harbour

- **Shelter** from waves and winds; calm water inside the harbour.
- **Adequate depth** for the largest ship (draught + under-keel clearance) and sufficiently wide, easily approached **entrance**.
- Good **holding ground** for anchors; enough area (water area for manoeuvring, turning basin

and berthing).

- Good connectivity by rail, road, inland waterways and pipeline to the hinterland.
- Navigation aids (lights, buoys), pilotage, tugs, dredging to keep depth; protection from siltation and littoral drift.
- Facilities: cargo handling equipment, storage, repair docks, fuel and water, medical, customs and security.
- Availability of land for expansion, local construction material, labour; economic viability with traffic potential.

2. Types of Ports

Basis	Types
Location	Sea port (coast), river / inland port, canal port, estuary port
Function / use	Commercial port (general, bulk, container), fishing port, naval (military) port, ferry port, port of refuge, oil / bunker port
Trade	Entrepôt / transhipment port (goods re-exported), ports of call (ships stop for fuel/ supplies), packet station
Ownership (India)	Major ports – under Central Govt. (e.g. Kolkata, Paradip, Visakhapatnam, Chennai, Kamarajar (Ennore), V.O. Chidambaranar, Cochin, New Mangalore, Mormugao, Mumbai, JNPT, Deendayal (Kandla)); minor ports – State

3. Classification of Harbours

UNIT – V : Ports and Harbours

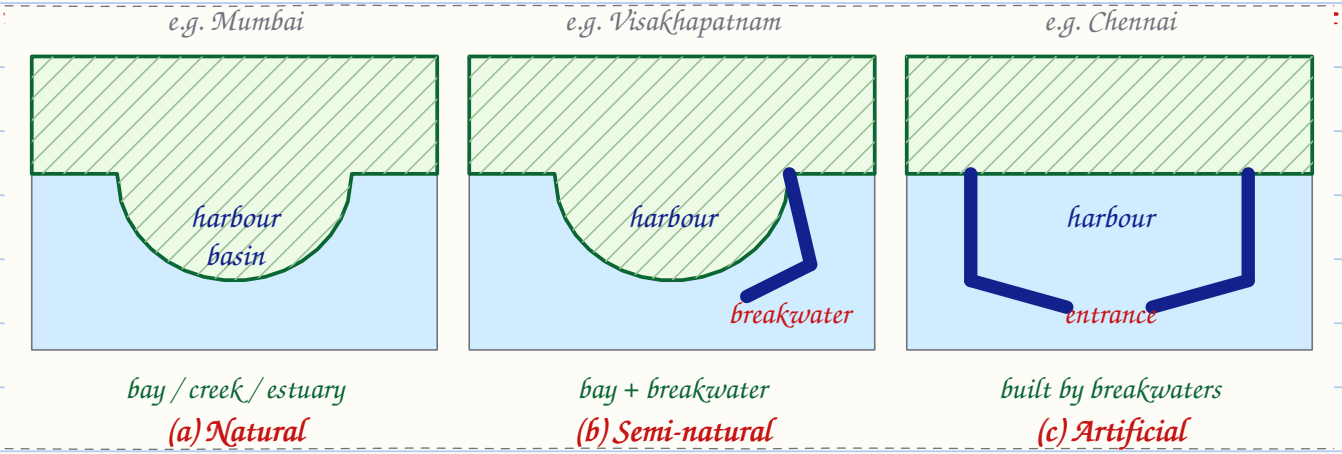


Fig: Natural, semi-natural and artificial harbours

(a) Based on the origin / protection

Class	Description	Remarks
Natural harbour	Formed by nature: bays, creeks, river mouths, estuaries; protected by headlands	Cheapest; limited by nature, e.g. Mumbai
Semi-natural harbour	Protected by nature on sides but opening to sea needs breakwater / jetties or dredging	e.g. Visakhapatnam
Artificial harbour	On an open coast; protection entirely by breakwaters, quay walls and dredged basin	Costly, e.g. Chennai

(b) Based on use / function

- Harbour of refuge – shelter for ships in storms (usually near busy sea routes, e.g. Dover); no trade facilities.
- Commercial harbour – for cargo and passengers; bulk, container, general cargo, oil terminals.
- Fishery harbour – for fishing boats, landing, auction hall, ice plant, repair; shallow depth.
- Naval / military harbour – for warships and submarines; secure, with dry docks and workshops.
- Inland (river) harbour – on rivers/lakes/canals for barges and smaller vessels.

Components of a harbour

- Entrance channel and approach; breakwaters; harbour basin (anchorage) and turning circle; quays, wharves, jetties (berths); docks; navigational aids; transit sheds and warehouses; roads and rail.

4. Tides, Waves and Littoral Drift (Basics)

- **Tides**: rise and fall of sea level by the gravitation of moon and sun. **Spring tides** (new/full moon) have the max range; **neap tides** (quarter moon) the min range. Tidal range decides the need for docks with lock gates.
- **Waves**: wind-generated; wave height H, period T, length $L = 1.56\,T^2$ (deep water, m); wave energy decides the design of breakwaters.
- **Littoral drift**: movement of sand along the shore due to waves hitting at an angle; may cause **siltation of harbour entrance** and erosion elsewhere → needs breakwater layout & dredging.
- **Draught**: depth of ship below water; **harbour depth** = draught + clearance (≈ 1 m or 10 %) + allowance for waves and siltation.

5. Breakwaters

Breakwater is a **barrier** built in the sea to **reduce wave action** and form calm, sheltered water in the harbour; it also protects the entrance from siltation, and may carry a berth or road.

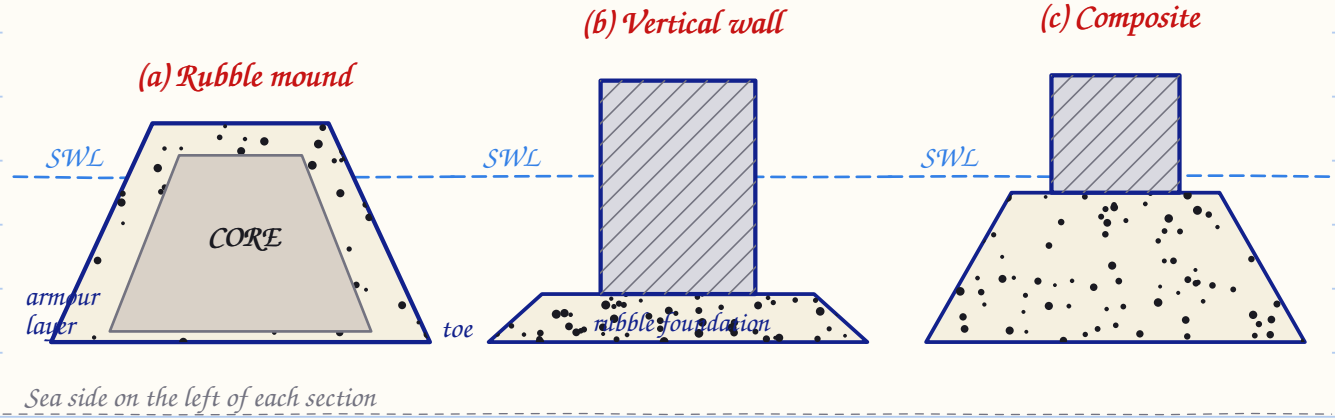


Fig: Types of breakwaters (sections)

UNIT – V : Ports and Harbours

Type	Description	Use
Rubble mound (heap)	Mass of quarried stone dumped with core of small stone and armour layer of heavy stone / concrete blocks (tetrapods, dolos, accropode); waves dissipate energy on the slopes	Most common; shallow to moderate water; flexible foundation
Vertical wall (monolithic)	Concrete blocks / caissons / masonry wall with vertical face on a rubble foundation; reflects waves	Deep water, firm bed; also used as quay
Composite	A low rubble mound topped by a vertical wall; combines both	Medium/deep water; economical
Horizontal composite	Vertical wall with a protective rubble berm / armour on the seaward side	Reduces wave reflection and impact
Special types	Floating (pontoons), pneumatic (air bubbles), hydraulic (water jets) – temporary or small harbours	Small, mild wave area

Layout of breakwaters

- Attached (shore-connected) – one or two arms from shore forming harbour entrance; e.g. Chennai, Visakhapatnam.
- Detached (offshore) – separate to the shore, parallel to coast; protects an open roadstead and reduces erosion.
- Orientation: the entrance is kept away from the direction of heavy waves and littoral drift; arms overlap slightly (staggered) to avoid direct waves entering.

Stability of rubble mound – Hudson formula

$$W = \gamma_r H^3 / [\, K_D (S_r - 1)^3 \cot\theta \,]$$

W = weight of armour unit (kN), γ_r = unit weight of stone (kN/m³), H = design wave height (m), $S_r = \gamma_r/\gamma_w$, θ = angle of slope, K_D = stability coefficient (≈ 4 for rough angular stone, non-breaking, trunk).

Example 1 – Armour stone

$H = 3$ m, $\gamma_r = 26$ kN/m³, $\gamma_w = 10.25$ kN/m³ (sea water), slope 2 H : 1 V ($\cot\theta = 2$), $K_D = 4$.

$S_r = 26 / 10.25 = 2.54$; $(S_r - 1)^3 = 1.54^3 = 3.65$.

$W = 26 \times 3^3 / (4 \times 3.65 \times 2) = 702 / 29.2 =$ 24 kN ≈ 2.4 tonne.

- For vertical wall breakwaters, wave force is found from Sainflou's method (non-breaking, standing wave) or Minikin's method (breaking wave); check sliding, overturning and bearing.

6. Docks and Types of Docks

A dock is an enclosed area of water (or a place) where ships are berthed, loaded/unloaded or repaired, kept at constant level, isolated from tidal changes.

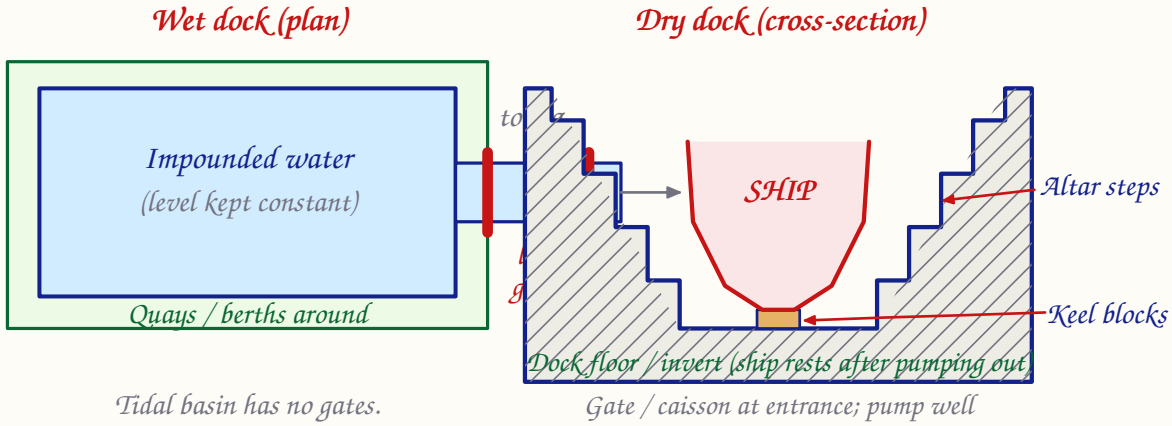


Fig: Wet dock (plan) and dry dock (section)

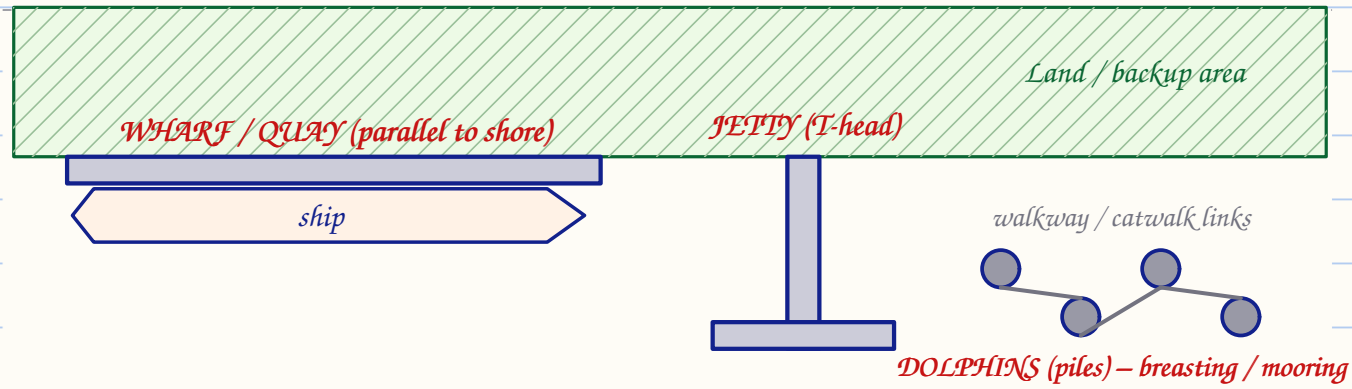
Type	Description
Tidal basin / open dock	Open to the tide; water level varies; no gates; for areas with a small tidal range
Wet dock (impounded dock)	Enclosed basin with lock gates; water kept at high-tide level; ships float at all times; for large tidal range (e.g. Kolkata, London). Entrance through lock with gates and sluices
Dry dock	Used to repair / inspect / paint the hull; can be pumped dry (details below)

Dry docks

- Graving dock (dry dock) – excavated basin with gate (caisson); ship enters at high tide, gate is closed, water is pumped out, and the ship settles on keel blocks; supported by shores; altar steps on sides give access for workers.

- Floating dock – a U-shaped pontoon, partially flooded to sink; ship enters; then water is pumped out and the dock rises with the ship; can be towed to other places.
- Marine railway / slipway (patent slip) – cradle on rails pulls the ship up an inclined slip; for small vessels. Lift dock: ship lifted by hydraulic platform.
- Components of graving dock: entrance, gate or caisson, dock floor (invert), side walls with altars, keel blocks, pumping station (pump well), culverts, capstans, crane and fenders.

7. Wharves, Quays and Jetties



Plan: berthing structures

Fig: Wharf, jetty and dolphins

Structure	Description
Wharf / quay	Structure parallel to the shore along which ships berth; backed by land for cargo handling. Quay = wharf of masonry / concrete wall
Jetty / pier	Structure projecting into the water (perpendicular or inclined to shore) so ships can berth at its sides; open (piles) so water flows; T-head or L-head jetty for tankers
Dolphin	Isolated pile cluster in water: breasting dolphin (takes berthing impact) and mooring dolphin (holds mooring ropes)
Fenders	Rubber/timber cushions on quay face absorbing berthing energy of ship

Types of wharf / quay walls

- Gravity wall (masonry / concrete blocks / caissons); sheet pile wall with anchors; cellular sheet pile; diaphragm wall; open piled deck (RCC deck on piles); relieving platform type.

- Quay face depth = draught + 1 m; width of apron 10-30 m; berth length ≈ ship length + 10 %; wharf crest level above highest tide + 1.5 m.

Berthing energy

$$E = \tfrac{1}{2} \cdot W \cdot V^2 / g \times Cm$$

W = displacement (kN), V = velocity of approach normal to quay (m/s), Cm = added mass coefficient ≈ 1.5 – 1.8. **Ex:** W = 50 000 kN, V = 0.15 m/s, Cm = 1.5 → E = 0.5 × 50000 × 0.0225 / 9.81 × 1.5 = **86 kN·m**.

8. Dredging Operations

Dredging is the process of **excavating** material (silt, sand, clay, rock) from the sea / river bed and transporting it for disposal, to **create** or **maintain** required depth.

- **Purposes**: capital dredging (new channel/basin), **maintenance dredging** (to remove siltation), reclamation (fill land), laying pipes/cables, sand mining, environmental clean-up.

Dredger	Working
Bucket ladder (bucket) dredger	Endless chain of buckets on a ladder digs and lifts material to hoppers; good for hard soils; stationary, anchored
Grab / clamshell dredger	Crane with a grab bucket; suits confined places, quay walls, deep digging
Dipper dredger	A power shovel on a barge; excavates hard soil, rocks and boulders
Suction dredger (plain suction)	Pump and suction pipe lowered to the bed; removes sand and silt as slurry
Cutter suction dredger (CSD)	Rotating cutter head loosens the soil which is sucked; pumps through floating pipeline to reclamation area; for hard soil, channel & land filling
Trailing suction hopper dredger (TSHD)	Self-propelled ship with drag-arm trailing along the bed; loads into **hopper**; sails to disposal area; for maintenance and long channels; no obstruction to traffic

- **Disposal**: dumping at designated sea area, pumping ashore for reclamation (beach nourishment), or use as fill. Environmental checks (turbidity, contaminants) needed.
- Dredging cost depends on soil type, quantity, distance to disposal, and depth; surveys with

echo sounder / bathymetry control depths.

9. Navigation Aids

Navigation aids help the master (captain) to find the position, safe channel and entrance of the port in day, night and fog.

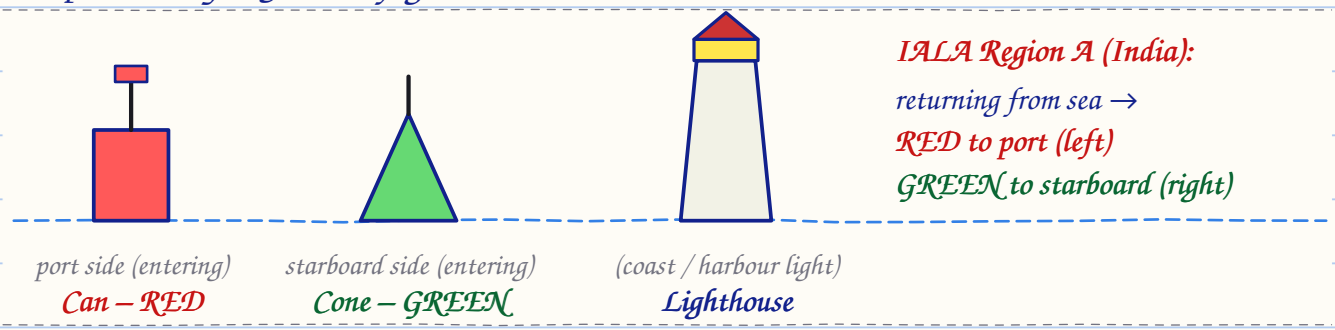


Fig: Navigation marks: buoys and lighthouse

Aid	Function
Lighthouse	Tall tower with a powerful rotating light for coast and harbour identification; visible 20+ nautical miles; each has a characteristic (flash pattern, colour)
Light vessel (lightship)	Anchored ship carrying a light and fog-signal where tower cannot be built (shoals, outer approaches)
Buoys	Floating marks anchored at the bed: lateral (can, cone, spherical, pillar, spar), cardinal, isolated danger, safe water and special marks; colours/shapes show side of channel
Beacons / leading (range) lights	Fixed marks on shore or piles; two lights in line (leading lights) show the centre line of the channel
Fog signals	Sirens, horns, bells, explosive signals, radio beacons used when visibility is poor
Electronic aids	Radar, GPS / DGPS, AIS (Automatic Identification System), VTS (Vessel Traffic Services), ECDIS electronic charts, echo sounder

- IALA Region A buoyage (India, Europe, Africa): entering from the sea, red marks on port (left), green on starboard (right) side.
- Other aids: pilotage, tugs, tide gauges, tide tables, harbour chart, signal stations, and weather warning systems.

10. Layouts of Harbours and Docks

- **Basic harbour layout**: sea → outer approach channel → entrance (between breakwater heads) → outer harbour / turning basin → inner basin with berths → docks / quays → transit sheds → rail & road.
- **Single-arm layout**: one breakwater built from the shore at the up-drift side of littoral drift; cheap, for mild climates.
- **Two-arm (converging) layout**: two arms form a narrow entrance; best protection; entrance oriented away from heavy waves; dredged entrance channel.
- **Dock layouts**: **finger (pier) type** – berths on both sides of parallel piers projecting from the shore; **parallel-quay (basin) type** – quays all round a basin; **marginal quay**; **tidal basin + impounded dock** arranged one behind other with a lock between.
- Width of a turning basin ≈ 2 – 2.5 × length of the largest ship (turning circle); width of entrance ≈ 1.5 – 3 × ship beam to length, so as to restrict waves but allow safe entry (practice varies).

Berth requirements

Item	Typical rule
Berth length	Length of design ship + 10 % (or about 15-30 m clearance for mooring)
Depth at berth	Maximum draught + 1 m under-keel clearance (more in soft bed)
Apron width	Cranes, rails, trucks: 20-40 m for container terminal
Quay crest level	Highest water level + freeboard (≈ 1 – 1.5 m)
Transit shed / yard	Close to the berth, rail and road access

11. Cargo Handling and Coastal Protection

Cargo handling

- **General cargo** by cranes and slings; **bulk cargo** (coal, ore, grain) by grab, conveyors, pneumatic unloaders; **liquid cargo** by pipelines at oil jetties; **containers** by gantry (ship-to-shore) cranes, yard cranes (RTG) and terminal tractors.

- Ro-Ro (roll-on roll-off) ramps for vehicles; passenger terminals for ferries; **Dry port / inland container depot (ICD)** as extension of the sea port.

Coastal protection works (related)

- **Groynes** – built perpendicular to shore to trap littoral drift; **seawalls / revetments** – protect shore from waves; **beach nourishment**; **offshore (detached) breakwaters** reduce wave energy and form tombolos.
- Siltation at harbour entrance is reduced by **sand bypassing** (pump sand from the up-drift side to the down-drift side).

12. Worked Examples

Example 2 – Depth of a harbour basin

Largest ship draught = 12.0 m; under-keel clearance = 1.0 m; allowance for waves and siltation = 0.5 m.

Depth required = 12.0 + 1.0 + 0.5 = **13.5 m** below the lowest water level (chart datum).

Example 3 – Wave length

Wave period T = 8 s. Deep-water wave length $L = 1.56\,T^2 = 1.56 \times 64 =$ **99.8 m ≈ 100 m**.

Example 4 – Berth length

Design ship length = 220 m. Berth length = 220 × 1.10 = **242 m ≈ 245 m**. For two berths in line: 220 + 220 + 3 × 15 m (spacing) = 485 m.

Example 5 – Hudson formula for a larger wave

H = 4 m, same stone and slope: $W = 26 \times 4^3 / (4 \times 3.65 \times 2) = 1664 / 29.2 =$ **57 kN ≈ 5.7 t** → weight increases with H^3 (here $(4/3)^3 = 2.37$ times).

Remember

- Wet dock – **lock gates** keep constant water level; dry dock – ship repaired after **pumping out**.
- Jetty ⊥ shore; wharf ∥ shore; dolphin = isolated piles.
- Capital dredging = new work; **maintenance dredging** = repeated silt removal.
- Breakwater weight of armour $\propto H^3$.

13. Summary and Quick Revision

Topic	Key points
Harbour	Sheltered water area; natural / semi-natural / artificial
Port	Harbour + facilities; sea, river, commercial, fishing, naval, refuge
Breakwater	Rubble mound, vertical wall, composite; Hudson: $W = \gamma r H^3 / [KD (Sr-1)^3 \cot \theta]$
Docks	Tidal basin, wet dock (lock gates), dry dock (graving, floating, slipway)
Wharf / Jetty	Wharf parallel to shore; jetty projects; dolphins for mooring
Dredging	Bucket, grab, dipper, suction, cutter suction, hopper dredgers
Navigation	Lighthouse, light ship, buoys (red port / green starboard), beacons, radar, GPS

Important Questions – Unit V

Short answer (2 marks)

- Differentiate between a port and a harbour.
- Name the types of harbours based on their origin.
- What is a breakwater? Write the Hudson formula.
- Define wet dock and dry dock; what is a jetty and a dolphin?
- What is dredging? Mention two navigation aids.

Long answer (10 marks)

- Explain the requirements of a good harbour and classify harbours with sketches.
- Explain the types of breakwaters with neat sketches and the layouts of harbours.
- What is a dock? Explain the types of docks with a neat sketch of a dry dock.
- Distinguish wharves and jetties; explain the types of quay walls and berthing structures.
- Explain the dredging operations and types of dredgers.
- Describe the navigation aids used in ports and harbours.

Quick revision: Port = harbour + facilities | natural / semi-natural / artificial | rubble mound, vertical wall, composite | tidal basin, wet dock, dry dock | wharf ∥ shore, jetty □

shore | red to port, green to starboard (IALA A).